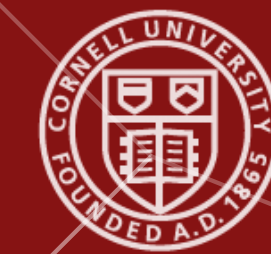


Temporal Abstraction in World Modeling for Long-Horizon Event Prediction

Haohuan Li Menghan Xu William Liaw Kevin Ellis
 hl2498@cornell.edu mx253@cornell.edu wjl86@cornell.edu kellis@cornell.edu
 Cornell Bowers Computer Science · BURE / AI-MI SURP



CornellBowers
Computer Science

Visual overview

Long-horizon prediction requires a world model that preserves useful information and decides how far ahead to reason. Temporal abstraction makes this tractable: latent steps span many timescales rather than a single tick. The Predictron learns a latent Markov reward process and blends predictions over several imagined depths, so the horizon is learned rather than fixed.

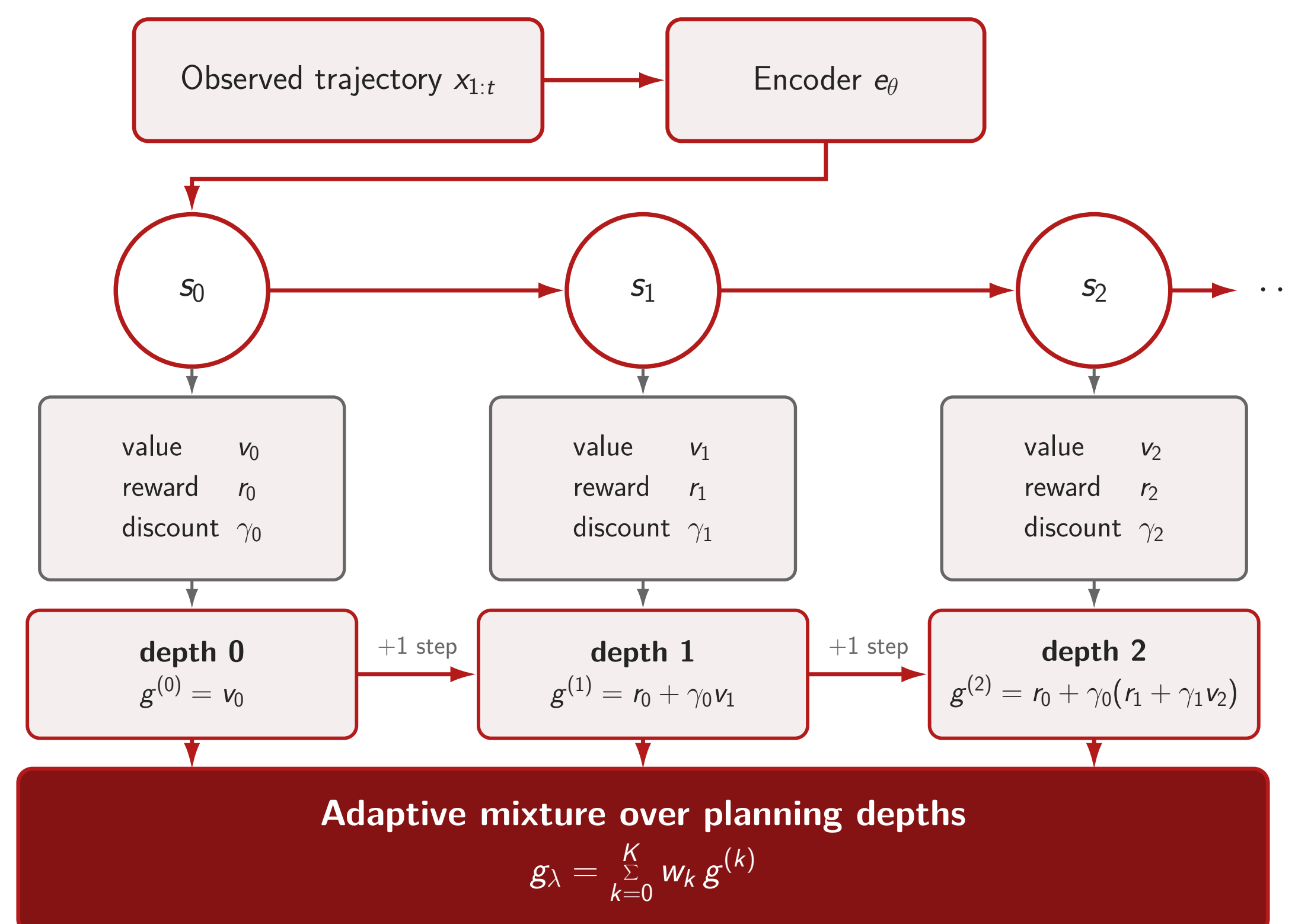


Figure 1: Latent planning architecture, adapted from Silver et al. [1].

Predictron

λ -Preturn:

$$g^{\lambda} = \sum_{k=0}^K w_k g^{(k)} \quad (1)$$

where:

$$w^k = \begin{cases} (1 - \lambda^k) \prod_{j=0}^{k-1} \lambda^j & \text{if } k < K \\ \prod_{j=0}^{K-1} \lambda^j & \text{if } k = K \end{cases} \quad (2)$$

Loss function

$$L^{0:K} = \frac{1}{2K} \sum_{k=0}^K \|\mathbb{E}_p[g | s] - \mathbb{E}_m[g^{(k)} | s]\|^2 \quad (3)$$

$$L^{\lambda} = \frac{1}{2} \|\mathbb{E}_p[g | s] - \mathbb{E}_m[g^{\lambda} | s]\|^2 \quad (4)$$

JESA: joint embedding successor architecture

JESA trains the predictron against a *separate* base encoder ϕ rather than against pixels: the model rolls its own latent forward one step and regresses onto $\phi(o_{t+1})$. The hatched arrow marks where the gradient is cut, which keeps the two branches from collapsing onto a constant embedding.

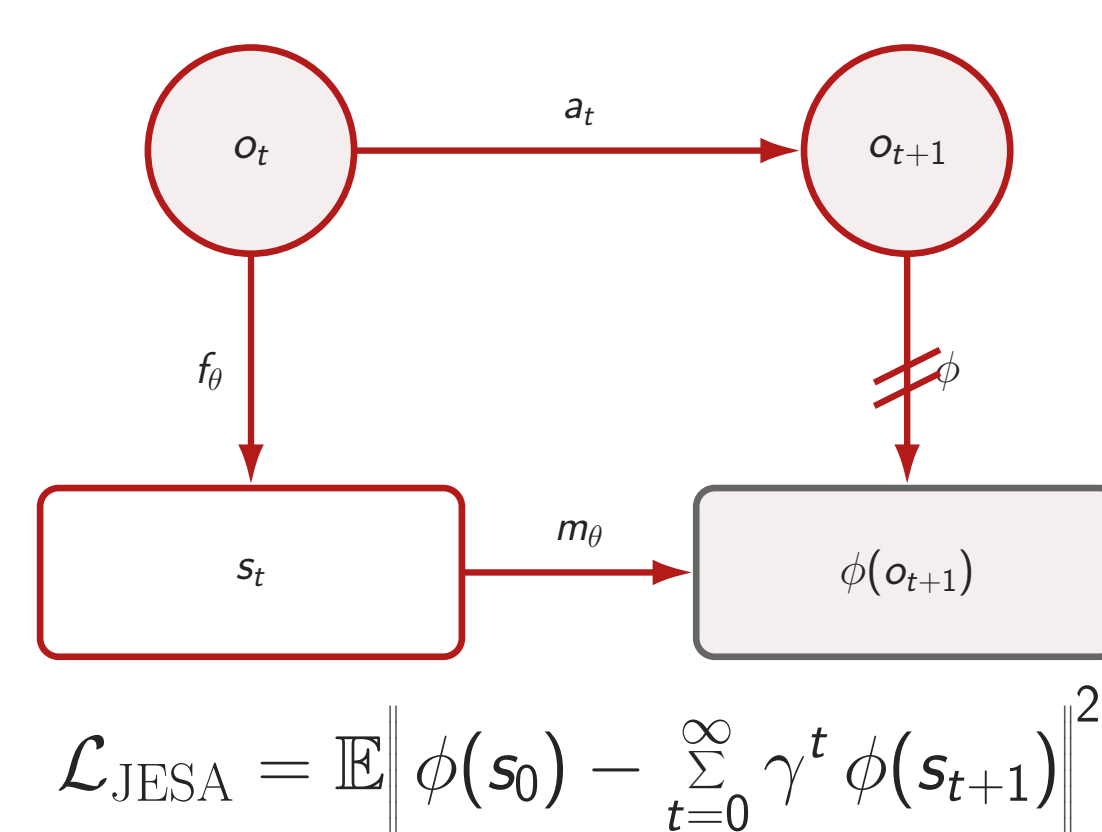


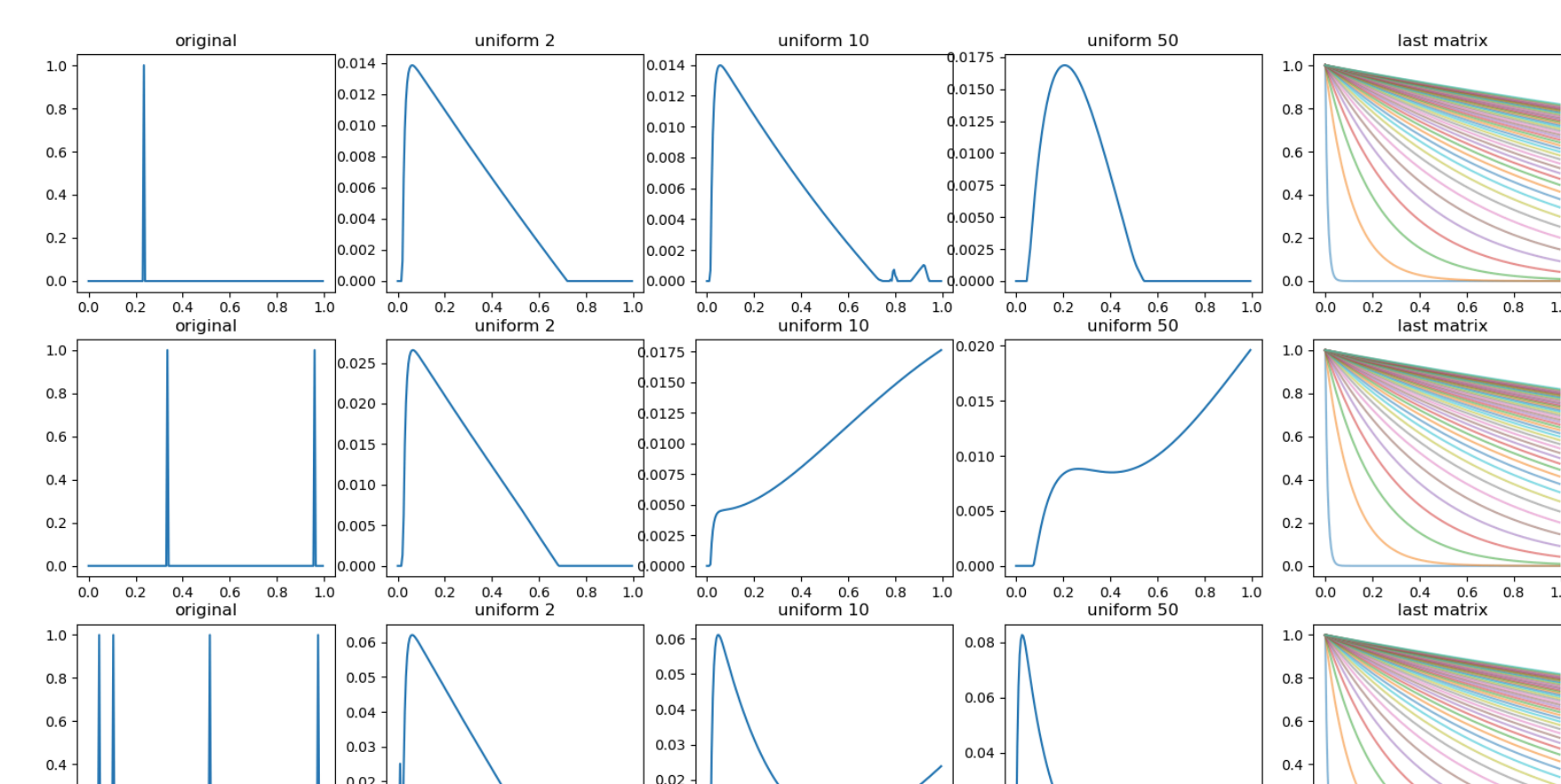
Figure 2: JESA as a commutative square; the hatched arrow is the stop-gradient.

Kernels and reconstruction

Written with a general kernel k , the return is

$$G_k = \int_0^{\infty} k(t) r(t) dt,$$

so the familiar discounted return is the exponential choice $k(t) = e^{-t/\tau}$. How faithfully the event train reconstructs from a *bank* of such returns depends on the kernel shape.



Introduction

Motivate the problem in 3–5 sentences. State why it matters and what gap your work fills.

- Key point one
- Key point two

Why joint embedding?

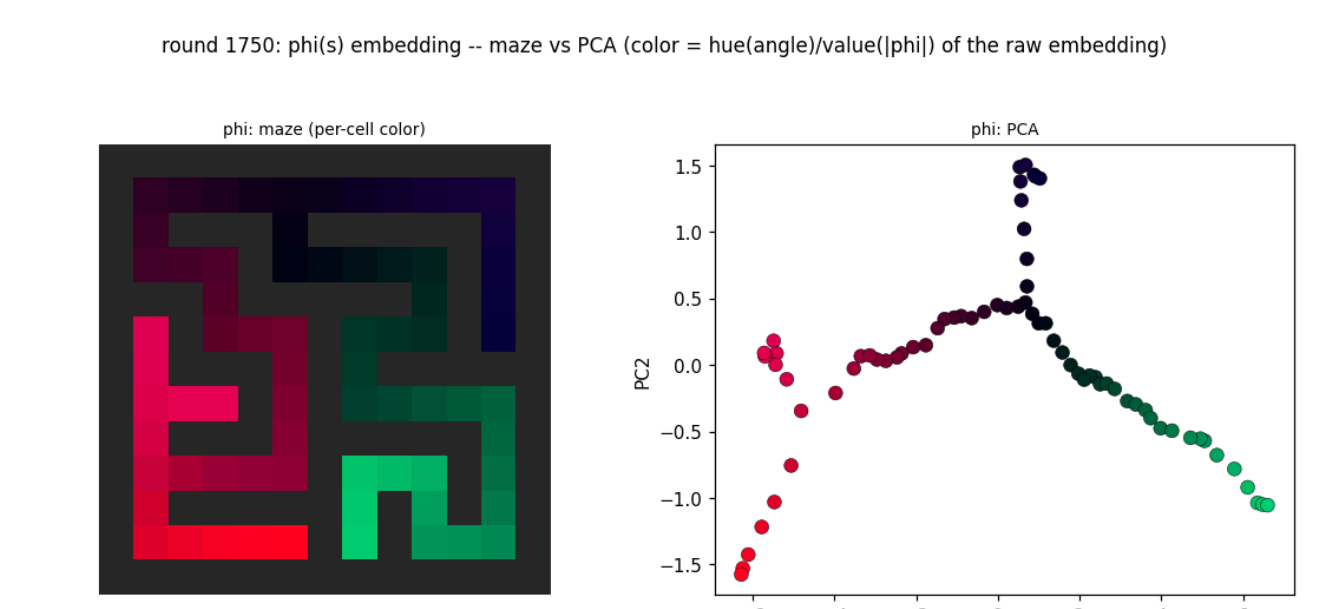


Figure 3: Base-encoder embeddings $\phi(s)$ over a maze (hue = embedding angle, value = $|\phi|$), with the PCA projection alongside.

ϕ is trained with **contrastive successor features** (CSF), the mutual information skill learning (MISL) objective of Zheng et al. [2]. MISL maximises the empowerment-style mutual information between states and skills,

$$I(s; z) = \underbrace{H(z)}_{\text{skill diversity}} - \underbrace{H(z | s)}_{\text{identifiability}},$$

which organises the embedding by long-horizon reachability rather than by appearance.

Successor features $\psi(s, z)$

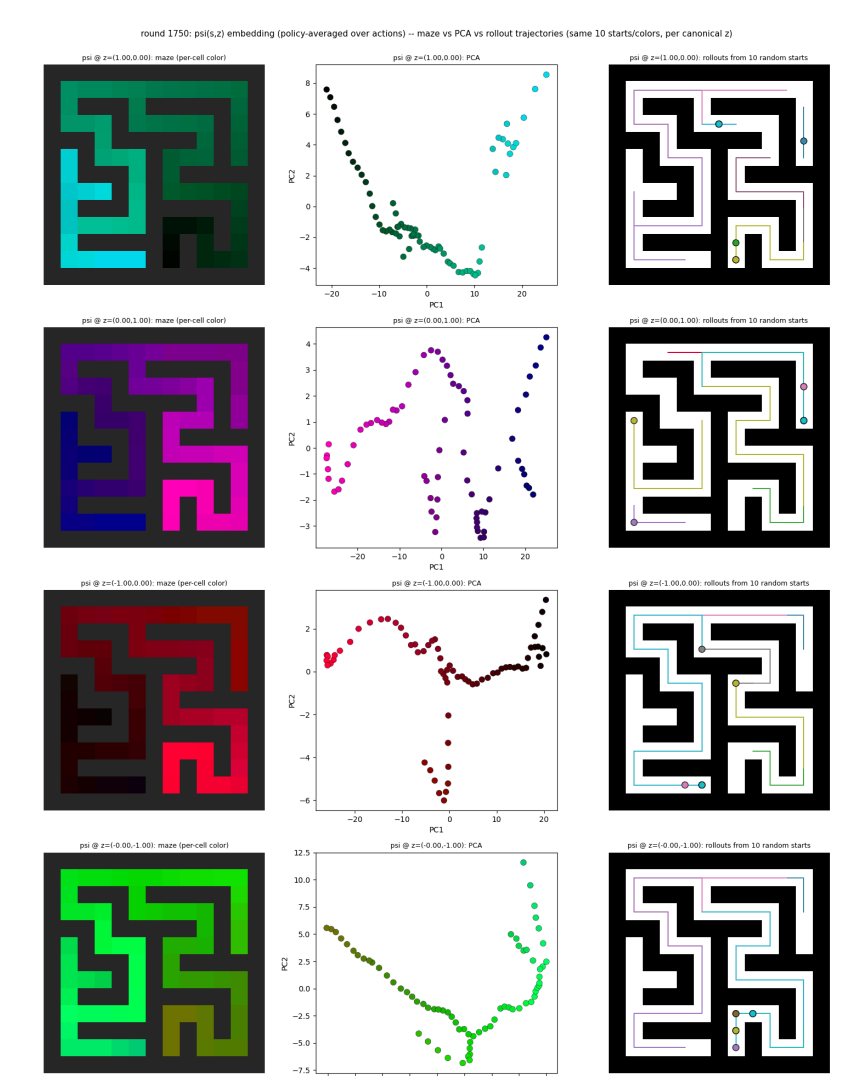


Figure 4: $\psi(s, z)$ per canonical z (rows): maze colouring, PCA, rollouts.

Each z induces its own field $\psi(s, z) = \mathbb{E}[\sum_k \gamma^k \phi(s_k) | s_0 = s, z]$: a fixed z traces a coherent long-horizon path.

Conclusions

- Main takeaway
- Limitation and future work

References

- [1] Silver, D., et al. (2017). The predictron: End-to-end learning and planning. *ICML*, 3191–3199.
- [2] Zheng, C., Tuyts, J., Peng, J., & Eysenbach, B. (2025). Can a MISL fly? Analysis and ingredients for mutual information skill learning. *ICLR*.